Generative AI Course Outline (6 Months) — Elite Developer Track

Course Overview

An advanced, build-heavy GenAI program for developers who want to become top-tier practitioners. The focus is depth, systems thinking, and production-grade skills across prompting, agents, RAG, fine-tuning, evaluation, safety, and deployment.

Target Audience

- Developers with solid Python and backend experience
- Comfortable with APIs, data pipelines, and cloud basics

Learning Outcomes

By the end of the course, learners will be able to: - Design and ship reliable GenAI systems, not just demos - Build multi-tool agents with robust error handling and observability - Implement high-quality RAG pipelines with reranking and grounding - Fine-tune and adapt models for domain performance gains - Evaluate GenAI systems with rigorous offline and online metrics - Apply security, safety, and governance controls in production - Optimize latency, cost, and UX for real products

Course Structure

- Duration: 6 months
- Format: Weekly lecture + lab + assignment
- Milestones: Monthly mini-projects + final capstone
- Expectation: Significant coding and independent research

Month 1: Core Foundations (No Handholding)

- Deep dive into LLM architectures and inference behavior
- Tokenization, context window trade-offs, and sampling mechanics
- Prompting as programming: constraints, schemas, and validation
- Reliability techniques: self-checks, fallback prompts, and confidence cues
- Mini-project: Prompt-driven assistant with strict schema output

Month 2: Tools, Agents, and RAG at Scale

- Function calling, tool routing, and state management
- Agent patterns (ReAct-style) with guardrails and planning
- Embeddings, vector stores, chunking, and metadata strategies

- Reranking, hybrid search, and citation grounding
- Mini-project: Production-grade RAG system with evaluation harness

Month 3: Adaptation and Fine-Tuning

- Decision framework: prompt vs RAG vs tuning
- Dataset design, labeling, and data versioning
- LoRA and parameter-efficient tuning at small scale
- Regression testing for tuned models
- Mini-project: Domain-tuned model with measurable gains

Month 4: Multimodal Systems

- Vision-language models and document intelligence pipelines
- OCR, layout understanding, and structured extraction
- Optional audio: speech-to-text and voice agents
- Multimodal orchestration and caching
- Mini-project: Multimodal workflow with structured outputs

Month 5: Evaluation, Safety, and Security

- Offline eval suites, LLM-as-judge, and human review loops
- Online evals, A/B testing, and feedback-driven iteration
- Prompt injection defense, data exfiltration prevention
- Governance, privacy, and compliance in production
- Mini-project: Red-team report + mitigation plan

Month 6: Deployment, MLOps, and Capstone

- Cost/latency optimization, routing, and caching strategies
- Monitoring, tracing, and incident response playbooks
- UX for GenAI: trust, controllability, and human-in-the-loop
- Capstone: End-to-end GenAI product with metrics and demo

Capstone Project (Suggested Ideas)

- Enterprise search assistant with hybrid retrieval and citations
- Analyst co-pilot with tools, workflows, and audit logs
- Multimodal document intelligence system

Assessment Breakdown

- Monthly mini-projects: 50%
- Mid-course check-in: 10%
- Capstone: 40%

Required Resources

- Python 3.10+
- Access to a GenAI API or local model
- GitHub repo for labs and assignments